

AI & ML for Data Scientists

Class 8: Classification and Risk Quant

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Introduction to Classification

Logistic Regression

Performance Measures

Multiclass Classification

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Multiclass Classification

What is Classification?

- Classification is the task of predicting a label (class).
- Binary classification: two classes (e.g., spam vs. ham).
- Multiclass classification: more than two classes.
- Multilabel and multioutput classifications

Applications of Classification Models

Finance:

- Credit scoring: Predict if a customer will default on a loan.
- Fraud detection: Classify whether a transaction is fraudulent.

Healthcare:

- Disease diagnosis: Predict presence of conditions (e.g., cancer, diabetes).
- Risk stratification: Identify high-risk patients.

Marketing:

- Customer segmentation.
- Predict likelihood to respond to a campaign.

LLM is also a Classification Model

- Why ?
- Word (token) prediction from the huge dictionary

Related Jobs: Risk Quants

- Credit Risk Quant: Models borrower defaults
- Market Risk Quant: Focuses on pricing derivatives and hedging strategies risk
- Operational Risk Quant: Analyzes tail-risk events (e.g., fraud, system failures)
- Model Risk Quant: model validation to prevent risks from modeling

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Why Linear Regression not for classification

- Why Linear Regression not for classification?
- Let me give you a example

Sigmoid function for the non-linearity

Equation 4-14. Logistic function

$$\sigma(t) = \frac{1}{1 + \exp(-t)}$$

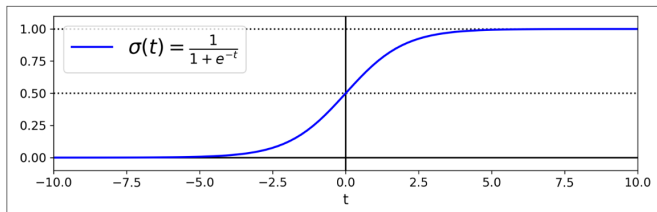


Figure 4-21. Logistic function

What is Logistic Regression?

- A statistical model for binary classification
- Predicts the probability of an event occurring
- Output ranges between 0 and 1
- Uses the logistic (sigmoid) function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Logistic Regression

- Traditional Logistic Regression:

$$\hat{P} = \sigma(x\theta^T) = \frac{1}{1 + e^{-x\theta^T}}$$

- Other machine learning models:

$$\hat{P} = \sigma(h_{\theta}(x)) = \frac{1}{1 + e^{-h_{\theta}(x)}}$$

- $\sigma(\cdot)$ is also called activation function (create non-linearity)
- $h_{\theta}(x)$ is also called score function, it can be ANN, Xgboost, random forest ...

From Soft Output ($\hat{P}$) to Hard Output ($\hat{Y}$)

Hard Output:

$$\hat{Y} = \begin{cases} 1, & \text{if } \hat{P} \geq \tau \\ 0, & \text{if } \hat{P} < \tau \end{cases} \quad \text{where } \tau \in [0, 1]$$

- $\hat{P}$ — soft output (probabilistic prediction)
- $\hat{Y}$ — hard output (binary class decision)
- τ — threshold (commonly set to 0.5)
- threshold low $\rightarrow$ more $\hat{Y}$ become 1
- Soft output $\hat{Y}$ is more important in finance (Why?)

Cross-entropy Loss Function (Cost function)

- Binary cross-entropy loss (log loss):

$$L(\theta) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(\hat{P}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{P}^{(i)})]$$

- Where:
 - m = number of training examples
 - $y^{(i)}$ = true label (0 or 1)
 - $\hat{P}^{(i)}$ = model predicted the probability, also called **soft output**

Cross-entropy Loss?

- Why we use Cross-entropy Loss?
- Bernoulli Distribution + MLE
- Pencil and paper time!!!

Optimization: Gradient Descent

- Iterative optimization algorithm
- Update rule:

$$\theta_{t+1} = \theta_t - \alpha \frac{\partial L(\theta)}{\partial \theta_t}$$

- Partial derivative:

$$\theta_{t+1} = \theta_t - \alpha \frac{\partial L(\theta)}{\partial \hat{P}(i)} \frac{\partial \sigma(h_{\theta}(x^{(i)}))}{\partial \theta}$$

Example: Training a Binary Classifier

- Example: Classify whether a digit is 5 or not.
- Model: `SGDClassifier` from Scikit-Learn.
- Uses stochastic gradient descent.

Measuring Performance

- Accuracy is not reliable for imbalanced datasets. (Why? Let me give a example)
- Confusion matrix is more informative.
- Precision and recall are better metrics.

Confusion Matrix

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Confusion Matrix on textbook (labels different)

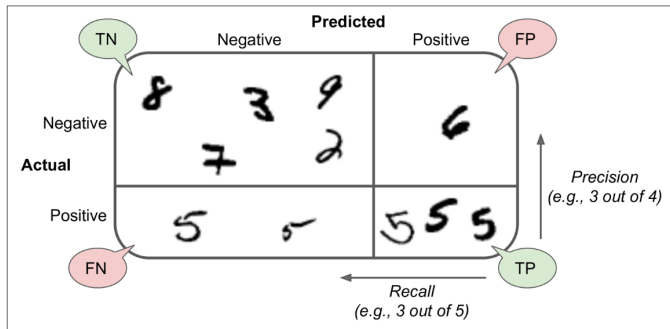


Figure 3-2. An illustrated confusion matrix shows examples of true negatives (top left), false positives (top right), false negatives (lower left), and true positives (lower right)

Precision vs Recall

- Precision = $\frac{TP}{TP+FP}$
- Recall = $\frac{TP}{TP+FN}$

Precision/Recall Trade-off

- You can adjust the decision threshold.
- $Precision(threshold)$ vs $Recall(threshold)$
- As we lower the threshold $\rightarrow$ Recall $\uparrow$ (we capture more true positives), But Precision $\downarrow$ (we also get more false positives).
- Higher precision $\Rightarrow$ lower recall and vice versa.

Precision/Recall Trade-off

Textbook (HandOn page 97): Predict Number 5 by change its threshold

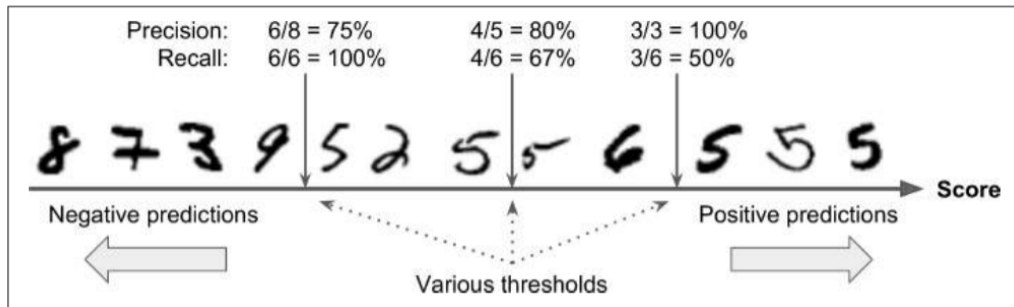


Figure 3-3. Decision threshold and precision/recall tradeoff

Precision/Recall Trade-off

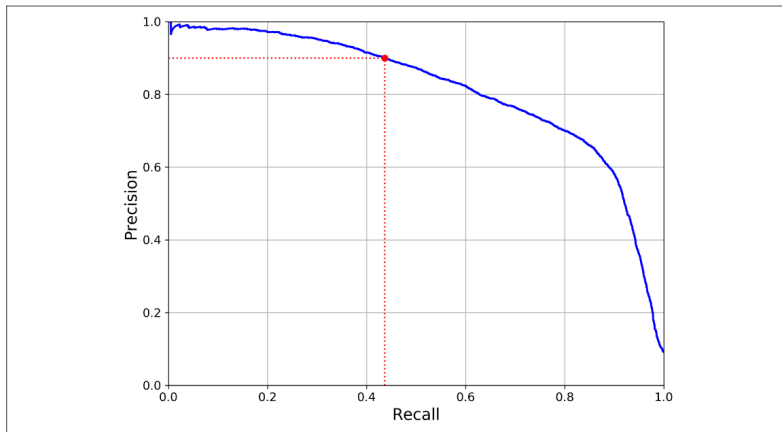


Figure 3-5. Precision versus recall

F1 Score

$$\begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix}$$

- Precision = $\frac{TP}{TP+FP}$
- Recall = $\frac{TP}{TP+FN}$
- F1 Score = $F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$ s.t. Harmonic mean of precision and recall

ROC Curve

- ROC = Receiver Operating Characteristic curve.
- Plots True Positive Rate (Recall) against False Positive Rate.
- $TPR = \frac{TP}{TP+FN}$
- $FPR = \frac{FP}{FP+TN}$
- Helps evaluate classifier performance at all thresholds.

ROC Curve

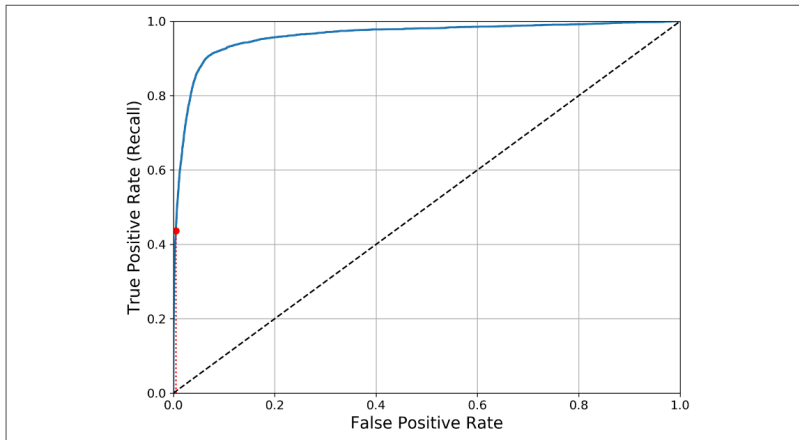


Figure 3-6. This ROC curve plots the false positive rate against the true positive rate for all possible thresholds; the red circle highlights the chosen ratio (at 43.68% recall)

ROC Curve: Compare models

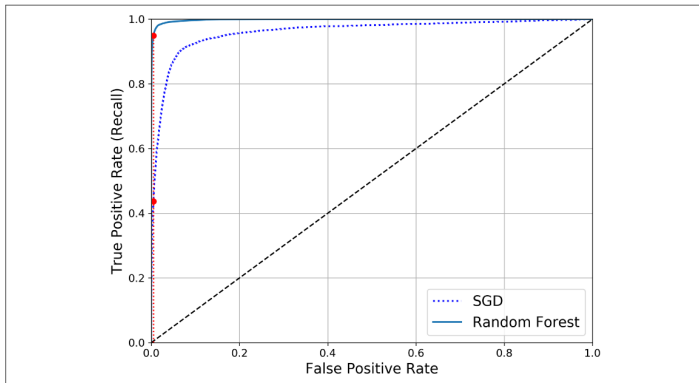


Figure 3-7. Comparing ROC curves: the Random Forest classifier is superior to the SGD classifier because its ROC curve is much closer to the top-left corner, and it has a greater AUC

Understanding AUC (Area Under Curve)

- AUC is the area under the ROC curve.
- $AUC = 1$: perfect classifier.
- $AUC = 0.5$: random guessing.
- The higher the AUC, the better the model distinguishes between classes.
- Useful when comparing multiple classifiers.

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Multiclass Classification: Softmax Function

Sigmoid vs Softmax

From Sigmoid → Softmax

Softmax Function

For a K -class classification problem:

$$\hat{p}_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}} \quad \text{where } k = 1, \dots, K \quad (1)$$

Cross-Entropy Loss (Softmax Loss)

Given the softmax output $\hat{p}$ and the true label y , the loss for **one observation** is:

$$\mathcal{L} = - \sum_{k=1}^K y_k \log(\hat{p}_k)$$

- **K is the number of classes**
- Penalizes incorrect predictions more when the predicted probability is low
- Encourages the correct class to have a high predicted probability

Reference

1. Hands-on Machine Learning with Scikit-Learn, Keras and TensorFlow (3rd edition)
2. <https://encord.com/glossary/confusion-matrix/>
3. Kaggle
4. Wikipedia
5. ChatGPT
6. DeepSeek